

* Practical - 8 *

-> Demonstrate supervised learning using artificial neural network.

```
from sklearn.neural_network import MLPClassifier
```

```
# Training Data
```

```
# X = studyhours
```

```
data_X = [[1], [2], [3], [4], [5], [6],  
           [7], [8], [9], [10]]
```

```
# Y = 0 -> Fail, 1 -> Pass
```

```
data_Y = [0, 0, 0, 0, 1, 1, 1, 1, 1, 1]
```

```
# Build Model with scikit-learn
```

```
model = MLPClassifier(hidden_layer_sizes = (8, 4),  
                       activation = 'relu',  
                       solver = 'adam',  
                       max_iter = 1000,  
                       random_state = 42)
```

```
# Train model
```

```
model.fit(data_X, data_Y)
```

Test Prediction

```
test_hours = [[2], [4], [6], [9]]  
predictions = model.predict(test_hours)
```

```
for i, hours in enumerate(test_hours):  
    print(f"Study Hours: {hours[0]},  
          Prediction: {predictions[i]}  
          (Pass = 1, Fail = 0)")
```

* Output

```
Study Hours: 2, Prediction: 0 (Pass=1, Fail=0)  
Study Hours: 4, Prediction: 0 (Pass=1, Fail=0)  
Study Hours: 6, Prediction: 1 (Pass=1, Fail=0)  
Study Hours: 9, Prediction: 1 (Pass=1, Fail=0)
```